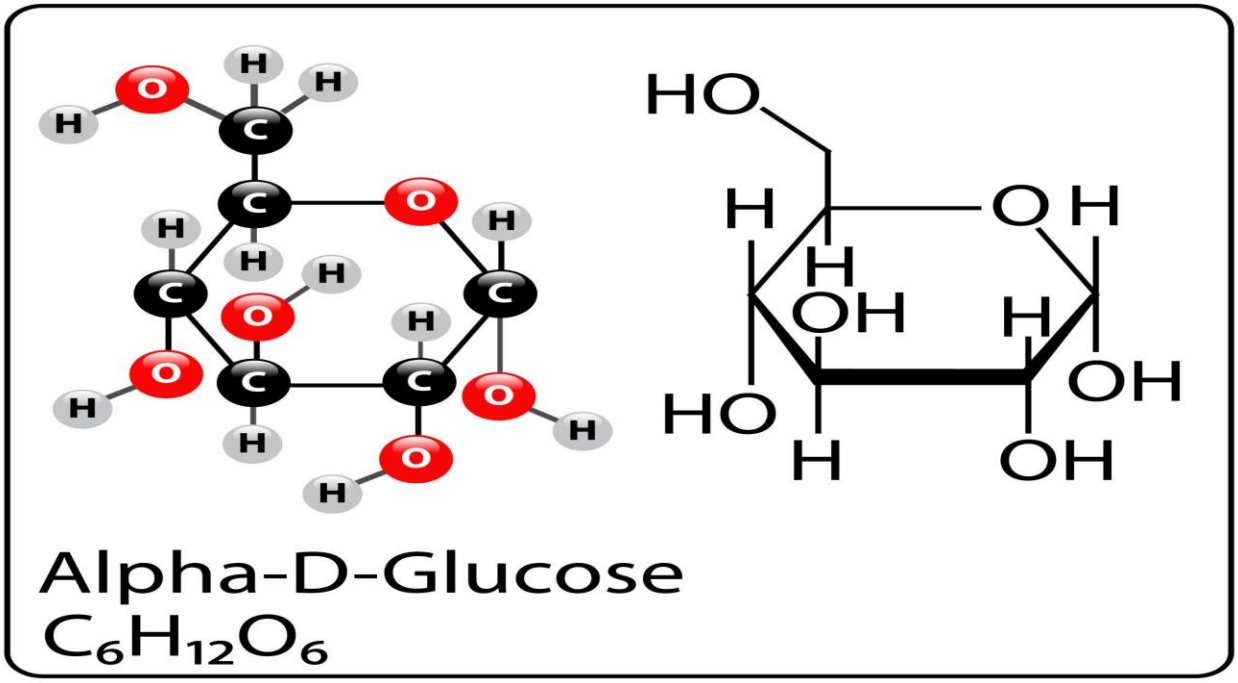
BIOMOLECULES

Introduction

Biomolecules are organic compounds present in living organisms that are essential for life processes. They include carbohydrates, proteins, lipids, nucleic acids, and vitamins. These molecules play structural, functional, and regulatory roles in the body. This chapter is highly important for NEET due to its conceptual and memory-based questions.

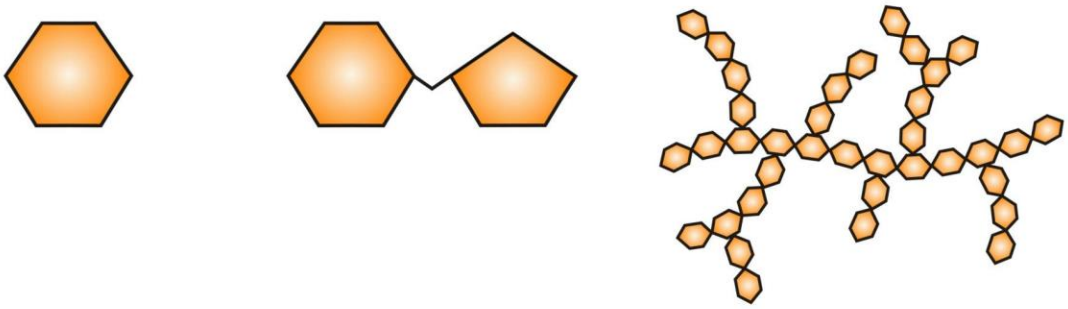
Carbohydrates

Carbohydrates are organic compounds composed of carbon, hydrogen, and oxygen. They are classified into monosaccharides, disaccharides, and polysaccharides based on the number of sugar units.



CARBOHYDRATES

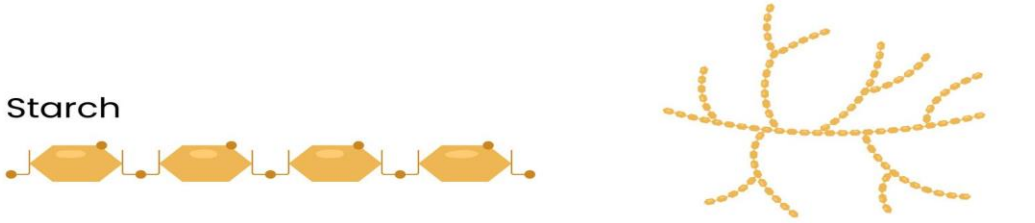
Monosaccharide **Disaccharide** **Polysaccharide**



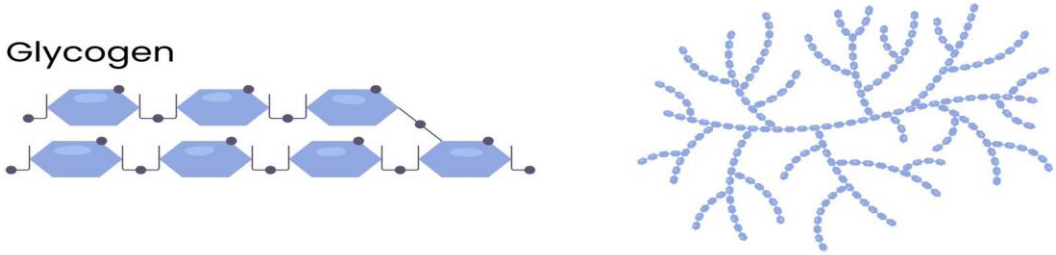
POLYSACCHARIDES

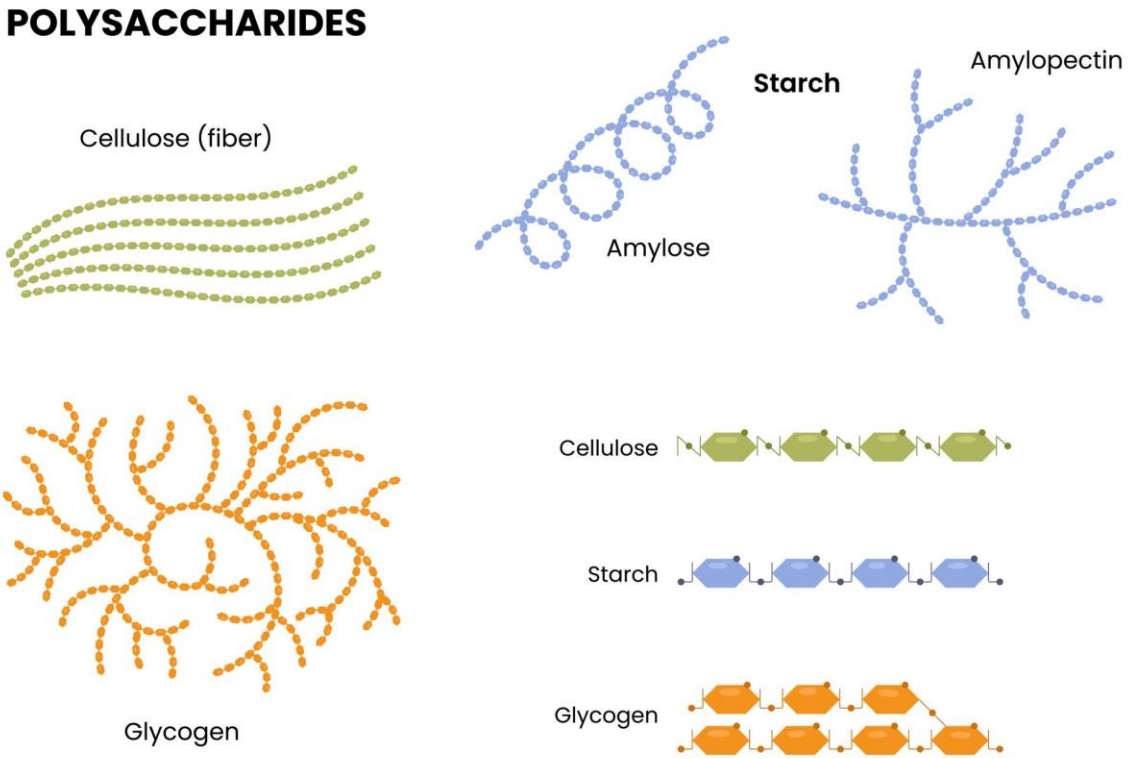
Cellulose

Starch



Glycogen

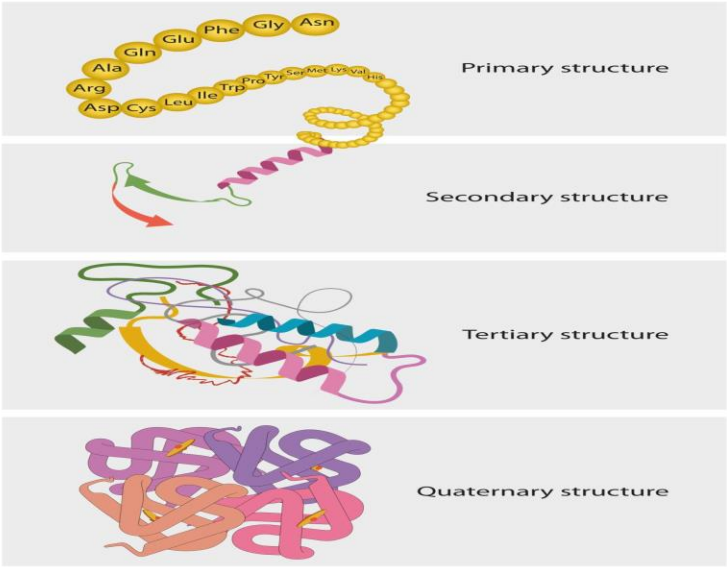




Monosaccharides like glucose are the simplest units, while polysaccharides like starch serve as storage forms.

Proteins

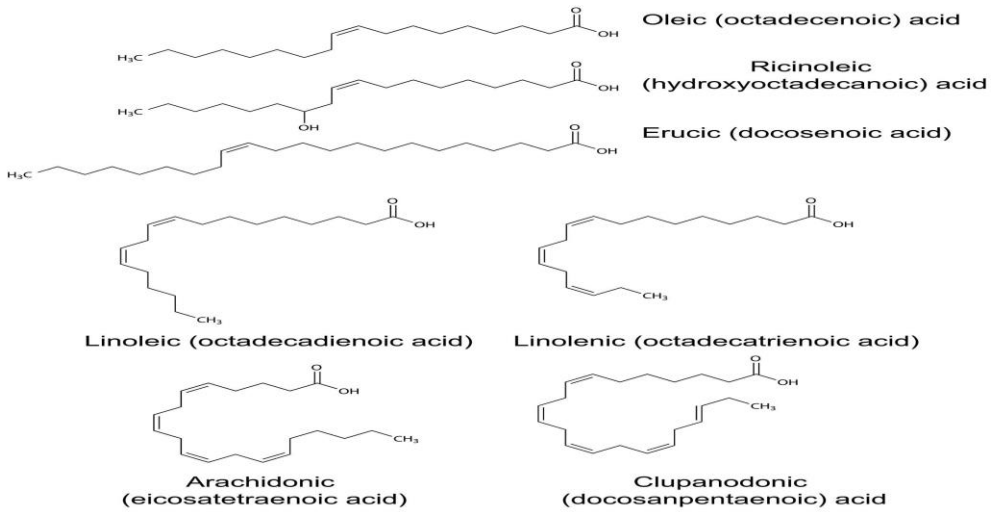
Proteins are polymers of amino acids linked by peptide bonds. They perform structural and functional roles such as enzymes, hormones, and antibodies.



Protein structure determines its function.

Lipids

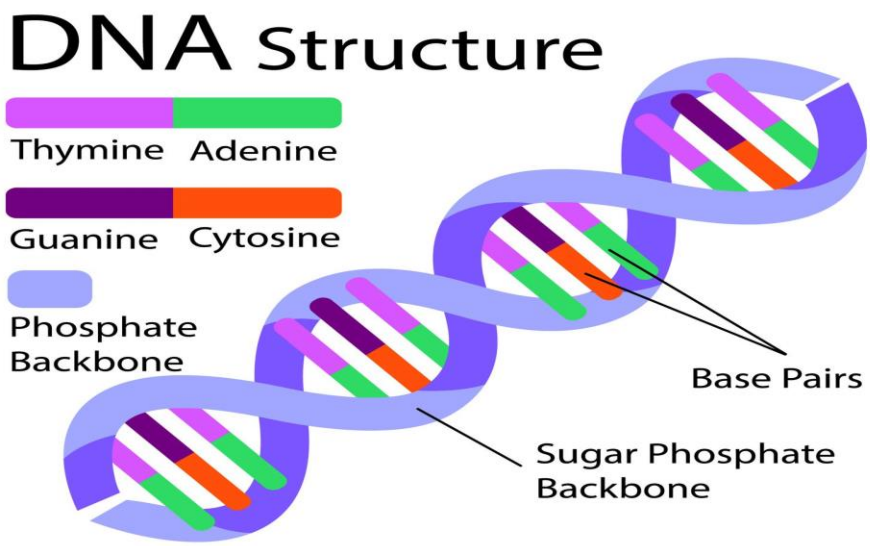
Lipids are hydrophobic molecules that include fats, oils, and waxes. They are composed of fatty acids and glycerol and serve as energy storage molecules.



They also form cell membranes and insulation layers.

Nucleic Acids

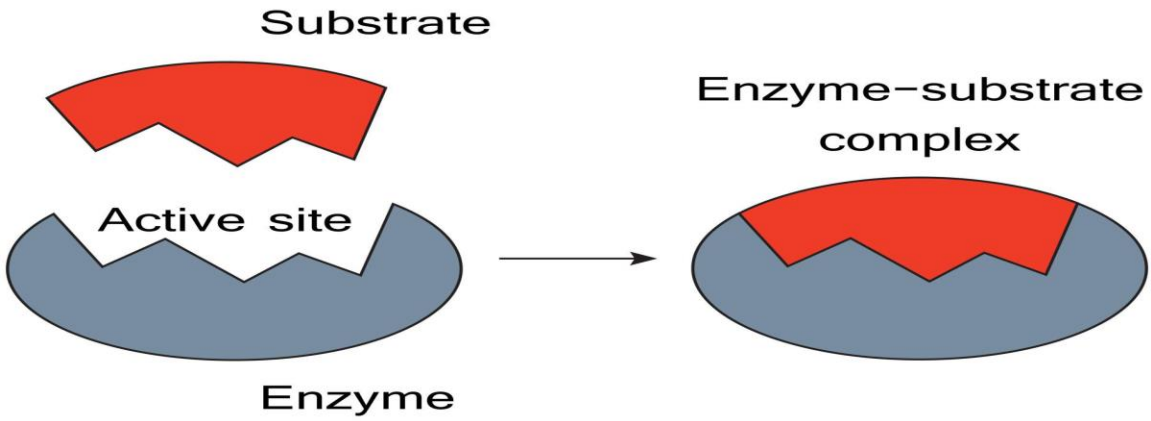
Nucleic acids are polymers of nucleotides and include DNA and RNA. They store and transmit genetic information.

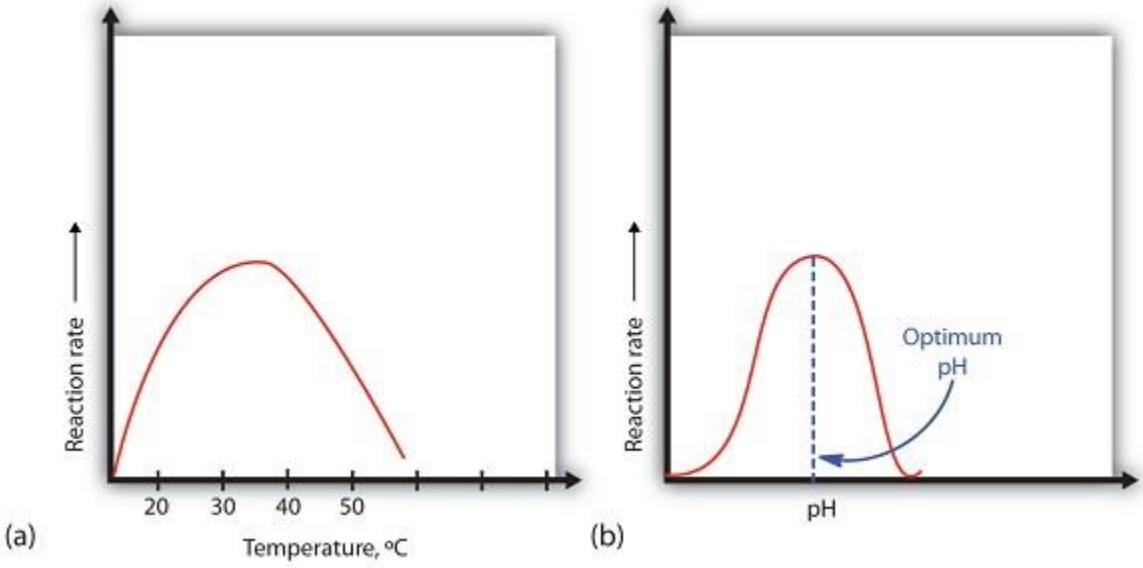


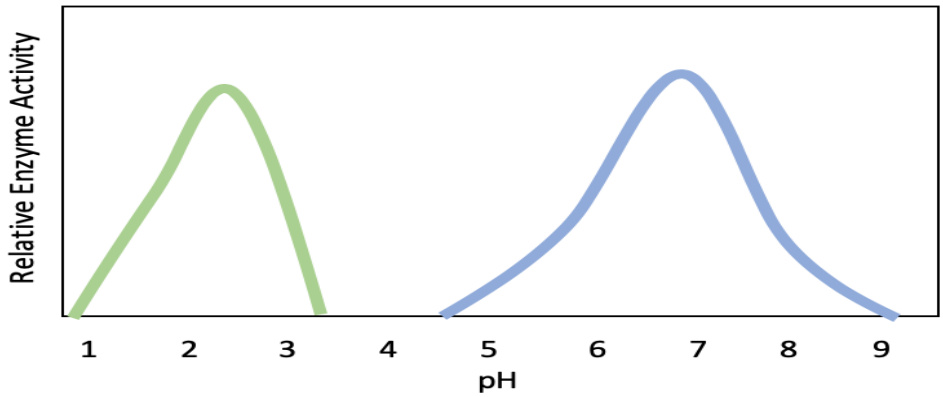
DNA contains genetic information, while RNA is involved in protein synthesis.

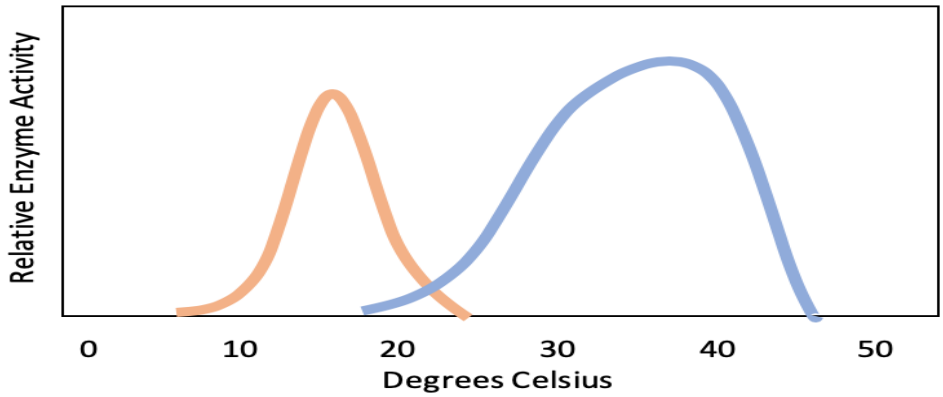
Enzymes

Enzymes are biological catalysts that speed up chemical reactions without being consumed. They are highly specific and function under optimal conditions.


Substrate
Active site
Enzyme
Enzyme–substrate complex


Reaction rate
20
30
40
50
Temperature, °C
(a)
Reaction rate
Optimum pH
pH
(b)

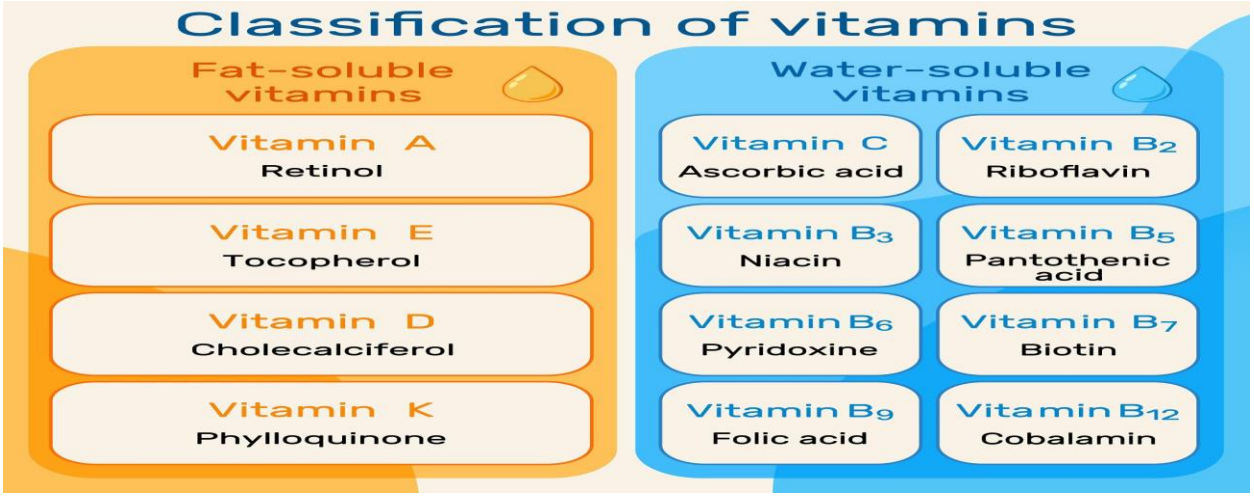




They lower activation energy and increase reaction rate.

Vitamins

Vitamins are organic compounds required in small amounts for normal metabolism. They are classified into fat-soluble and water-soluble vitamins.



Deficiency of vitamins leads to various diseases.

Hormones

Hormones are chemical messengers that regulate physiological processes in the body. They are secreted by endocrine glands and act on target organs.

Metabolism

Metabolism includes all chemical reactions occurring in living organisms. It consists of anabolic reactions that build molecules and catabolic reactions that break down molecules.

Conclusion

Biomolecules are essential for life and play a crucial role in maintaining biological functions. Understanding carbohydrates, proteins, nucleic acids, and enzymes is important for NEET preparation. This chapter is highly scoring due to its conceptual clarity and diagram-based questions.